Botulinum Toxin A Adjunctive Use in Manual Chemabrasion: Controlled Long-Term Study for Treatment of Upper Perioral Vertical Wrinkles

Bogdana Victoria Kadunc, PhD, Ada Regina Trindade de Almeida, MD, Adriana Amorin Vanti, MD, and Nilton di Chiacchio, PhD*

BACKGROUND The treatment of perioral vertical wrinkles is a very common request from female patients as of their 40th decade of life. Actinic damage, cigarette smoking, loss of deep structures volume, sleep positions, orthodontic deformities, and dynamic components have been thought to cause this aesthetic problem.

OBJECTIVE The objective was to investigate the combination of botulinum toxin type A (BoNTA) pretreatment of the orbicularis oris muscle with manual chemabrasion to increase its clinical efficacy in the therapy of upper perioral vertical rhytides.

METHODS Twelve women presenting moderate to severe upper lip vertical rhytides were randomized to receive two unilateral BoNTA injections at the vermilion border, 1 week before chemabrasion (35% trichloroacetic acid followed by dermasanding) of the upper perioral cosmetic unit. Wrinkle severity was assessed by two blinded observers at baseline, 30 days, 90 days, 180 days, and 3 years by using a four-point Facial Wrinkle Severity Scale (FWSS).

RESULTS From Day 90 to Year 3, the BoNTA-injected sides showed smaller grades in the FWSS than control sides ($p < .05$).

CONCLUSION Previous injections of BoNTA improve short- and long-term results of chemabrasion in the upper lip region.

Ada Almedia, MD, is a consultant and speaker for Allergan, South America.

The perioral aging process has been treated by using different modalities: ablative techniques, surgical approaches, botulinum toxin type A (BoNTA) injections, and fillers.[1] Severe static rhytides are traditionally treated with resurfacing procedures.

Chemodenervation through BoNTA has proven to be safe and effective in the treatment of a number of disorders, including hyperkinetic facial lines in the upper third of the face. Its use in the mid- and lower face, particularly in the perioral region, has been increasing for aesthetic treatment; however, it is crucial that very low dosages and precise injection sites are used to avoid impairment of the orbicularis oris muscle sphincter function.[2]

Phenol chemical peeling, dermabrasion, and laser resurfacing are ablative techniques showing similar results, depending on the physician's skills. Chemabrasion combines chemical peel with dermabrasion, and it is a useful technique to be performed in the perioral region.[3]

Minimized tension on wound edges during healing phase knowingly favors final cosmetic results.[4] The use of BoNTA to reduce the dermabraded perioral area mobility would provide a short healing period and better final results.

The purpose of this study was to investigate whether previously treating the orbicularis oris muscle with BoNTA injections enhances manual chemabrasion

*All authors are affiliated with the Dermatologic Clinic, Hospital do Servidor Publico Municipal, São Paulo, Brazil

© 2007 by the American Society for Dermatologic Surgery, Inc. • Published by Blackwell Publishing •
ISSN: 1076-0512 • Dermatol Surg 2007;33:1066–1072 • DOI: 10.1111/j.1524-4725.2007.33220.x

TABLE 1. Patients' Characteristics and Adverse Effects

Patient	*Age*	*Phototype*	*Initial FWSS score*	*Day 9*	*Day 17*	*Day 30*	*Day 90*	*Day 180*	*Year 3*
MICD	65	IV	3	E	HS	PIH			
SCO	63	II	3	E	EDD	ER			
LET	62	III	3	E	EDD	ER			
SSM	59	III	3	E		PIH			
LRC	69	III	2	E		ER			
AFT	57	II	3	E	EDD	ER	ER/LH	LH	LH
SFB	47	IV	2	E		PIH			
MCM	60	III	3	E		ER			
IW	62	II	3	E	EDD	ER			
MS	55	III	2	E		ER			
MCL	50	II	2	E		ER			
IC	64	III	3	E		PIH			

E, hemilip edema; HS, herpes simplex; EDD, eating and drinking difficulty; ER, erythema; PIH, postinflammatory hyperpigmentation; LH, localized hypochromia.

clinical efficacy when treating upper perioral vertical rhytides.

Methods

Patients

Twelve women aged between 47 and 69 years were selected to participate in this prospective, randomized, double-blind, intrapatient controlled study, where each patient served as her own control.

Upper perioral vertical rhytides were assessed with a four-point facial wrinkle severity scale (FWSS) where 0 = no wrinkles; Grade 1 = mild; Grade 2 = moderate; and Grade 3 = severe wrinkles. Eight of 12 patients had her rhytides classified as severe (Grade 3) and 4 as moderate (Grade 2). Four patients were classified as Fitzpatrick phototype II, 6 as III, and 2 as IV (Table 1).

Exclusion criteria were any other cosmetic procedures or isotretinoin intake within the previous 12 months; anterior deep ablation in the perioral area; history of herpes simplex flares, neuromuscular diseases, and allergy to human albumin or BoNTA; wind instrument musicians or professional speakers; pregnancy; or lactation.

An informed written consent was obtained from all patients. The study protocol conformed to the guidelines of the 1975 Declaration of Helsinki and was approved by our institutional review board. Patients were educated about the expected transient asymmetry that would occur during the study. High-quality standardized photographs were obtained from frontal and two lateral views, at rest and during maximum orbicularis oris muscle contraction, at baseline, and at each follow-up visit.

Botulinum Toxin

Just before use, BoNTA (BOTOX, Allergan Inc., Irvine, CA) was reconstituted in 2 mL of saline solution, resulting in a concentration of 50 µm/mL. Patients were randomized to receive BoNTA injections into only one side of the upper lip. For the randomized side, two sites were injected with 1.5 units at the vermilion border in the middle of each quadrant. The other side was injected under the same conditions, but this time with saline solution. This was considered Day 0.

Chemabrasion

Prophylactic 2,400 mg/day acyclovir doses were given before and after chemabrasion.[5] This procedure was performed 7 days after the BoNTA injection



Figure 1. Hemiorbicularis contraction: 30 days after BoNTA injections into the right side and 21 days after chemabrasion.

(Day 7). Initially, a 30% trichloroacetic acid (TCA) peeling was performed on the face, except for the upper lip area. Afterward, an intraoral infraorbital nerve block anesthesia with 2% lidocaine and epinephrine was used, before applying 35% TCA on the upper lip area. Immediately after the establishment of uniform white frosting, manual dermasanding was performed utilizing No. 100 and 220 (Norton, São Paulo, Brazil) wet sandpaper pieces wrapped in dental anesthesia tubes. The abrasion end point was the onset of abundant and confluent bleeding dots and visualization of sebaceous glands, thereby characterizing the presence of upper to mid-reticularis dermis. All patients were treated under the same conditions by the same author to avoid technique variations.

Postoperatively, semiocclusive dressings consisting of sterilized veil fabric[6] and 2% fusidic acid cream (Verutex cream, Roche, Basel, Switzerland) were used for 48 hours. Immediate follow-up visits occurred 2 (Day 9) and 10 (Day 17) days after chemabrasion to evaluate the grade of edema, acute complications, and reepithelialization rate. On Day 30, possible postoperative erythema, postinflammatory hyperpigmentation, and residual static or dynamic rhytides were assessed.

An intermediate follow-up visit occurred on Day 90 to evaluate late pigmentation phenomena and static or dynamic rhytides. Late follow-up visits occurred on Day 180 and Year 3 to evaluate clinical improvement level and maintenance of results. They did not receive any other BoNTA treatment within this period.

Assessment

Patients were clinically evaluated in blinding (Observer 1) and by comparison through pre- and post-treatment photographs (Observer 2) by using the FWSS. Scores were obtained at baseline and each follow-up visit.

The statistical analysis was performed using the nonparametric Wilcoxon signed-rank test. A p value of less than .05 was considered significant.

Results

All patients completed the 36-month follow-up period. Seven days after BoNTA injections, all treated sides presented clear decrease of muscular motion with clear deviation of the lip midpoint, resulting in a hemiorbicularis contraction (Figure 1).

On Day 9, edema was greater on the weakened side in all patients. On Day 17, all of them completed the reepithelialization. Four individuals (33.3%) complained about interference with eating and drinking and one (8.3%) experienced a mild herpes simplex eruption, in spite of prophylaxis.

On Day 30, eight patients (66.6%) exhibited several grades of erythema, similar on both lips sides, which gradually disappeared. The two patients classified as Fitzpatrick phototype IV and the two classified as type III (33.3%) showed transient postinflammatory hyperpigmentation, which was resolved with standard bleaching agents (Table 1). The BoNTA-injected hemilips showed eversion and fullness, mainly on the medial quadrant (Figure 2).

The difference of power contraction between the two sides was no longer perceptible at the 90-day follow-up visit (Figure 3). In six individuals, with long



Figure 2. Enhanced hemilip eversion and fullness: 30 days after BoNTA injections into the right side and 21 days after chemabrasion.

rhytides, the effacement was evident near the vermilion border, remaining insufficient on the superior part near the nose. One patient exhibited a localized hypochromic patch at the control side.

There was no statistical difference between the two observers' scores ($p > .05$). The analysis between the injected BoNTA and the control sides was performed by using the data of Observer 1. At baseline, the scores were the same for both sides. After 6 months, 10 of 12 patients (83.3%) presented smaller scores in the FWSS on the BoNTA-injected sides. The score decreased: one grade for 2 patients (16.6%); two grades for 4 patients (33.3%); and three grades for 4 patients. One (8.3%) patient did not change her grade at all, and another improved similarly on both sides. After 3 years, scores were maintained ($p < .05$; Table 2).



Figure 3. 90 days after BoNTA injections into the right side —symmetrical orbicularis contraction.

Discussion

Loss of volume, dermatoheliosis, and dynamic action are the main etiologic causes of vertical perioral rhytides, also known as smoker's lines. BoNTA is the gold standard treatment for glabellar, forehead, and periocular dynamic wrinkles and can be an excellent adjunctive method, maintaining the effect of other procedures.[2,7–9]

According to the literature, conservative doses of BoNTA causing "microparesis of the orbicularis oris muscle"[2] might reduce the dynamic component of lip aging, resulting in rhytides amelioration, upper lip border eversion, and increased lip volume.[10]

The recommended technique is limited amounts of toxin and very precise injections away from midline and mouth corners to prevent flat and/or incompetent lips.[2,10,11] The imperative use of low doses, instead of effacement, will only result in an amelioration of moderate to severe vertical upper lip rhytides[10] in patients who are exclusively treated with BoNTA. When the perioral rhytides are nondistensible, multiple, deep, and localized in an elastotic skin, the use of a resurfacing technique is mandatory to achieve remodeling of tissues and reorganization of the collagen and elastic fibers.

Comparative studies of different perioral ablative methods are found in the literature: CO_2 laser versus motor[12] or manual dermabrasion[13,14] or Baker-phenol chemical peeling.[15] Phenol chemical peeling showed, since the 1960s, the best performance, although causing hypopigmentation related to its deep action in the mid-reticularis dermis.[16] Dermabrasion allows more controllable planning of the area of concern, reducing the risk of undesirable deepening of the abrasion, and can be conducted in two different ways: by motor device equipped with diamond fraises or brushes or manually with sandpaper.

TABLE 2. FWSS Scores in BoNTA-Injected and Controlled Sites at Baseline, Day 30, Day 90, and Day 180, and Year 3

	Number of patients									
	Baseline		*Day 30*		*Day 90*		*Day 180*		*Year 3*	
FWSS scores	*BoNTA*	*Control*	*BoNTA*	*Control*	*BoNTA*	*Control*	*BoNTA*	*Control*	*BoNTA*	*Control*
0	0	0	2	1	6	1	6	1	6	1
1	0	0	7	3	5	4	4	3	4	3
2	4	4	3	8	1	7	2	8	2	7
3	8	8	0	0	0	0	0	0	0	1
p Value	1.000		.031		.002		.004		.002	

BoNTA, botulinum toxin type A.

The latter is considered the method of choice by Chiarello.[14]

The chemabrasion technique is the association of dermabrasion and chemical peeling. Stagnone[3] described this technique in 1977, utilizing 88% phenol followed by motor dermabrasion, and this was popularized by Harris and Noodleman[17] and Lusthaus and coworkers,[18] by using sandpaper.

We chose to treat the perioral unit of our patients with 35% TCA followed by dermasanding. It is very useful and quick to perform, requiring low material costs. Skin ablation is facilitated by necrosis due to TCA action, while the surgeon and operating room personnel are protected from aerosolized blood and skin fragments. We have been using this type of chemabrasion, in different facial cosmetic units, for the past 10 years with good results and safety.[19]

It is very important to blend abraded and non-abraded regions, reducing demarcation lines, mainly at the periphery of the treated unit and neck area. For this reason, a 30% TCA peel was performed in other facial regions.

In our study, at the 48-hours follow-up visit, a greater grade of edema was observed on the BoNTA-treated upper lip half. It may be explained by the impaired drainage caused by the relative immobility of the treated side. After 30 days, BoNTA injected hemilips showed diminished power contraction, noticeable eversion and lip fullness, mainly in the medial quadrant. The difference of dynamic effects between the two sides in our cases completely disappeared at 90 days, and this finding is similar to that encountered in other articles.[10,11]



Figure 4. Patient before BoNTA injections into the right side.



Figure 5. Patient 6 months after BoNTA injections into the right side.



Figure 6. Patient 3 years after BoNTA injections into the right side.

Other authors described the maintenance of results when combining BoNTA and CO_2 laser resurfacing.[7,20,21] West and Alster[20] observed better and long-lasting effects in 20 patients treated with BoNTA after resurfacing, when compared to controls.

Zimbler and colleagues[21] compared, in a prospective, randomized, blinded study, facial laser resurfacing with and without BoNTA pretreatment of upper third facial lines. They concluded that the hyperdynamic facial lines heal in a smoother rhytide-diminished fashion if pretreated with BoNTA injections.[21]

The explanation for these findings is that BoNTA injections minimize the recurrent muscle activity that can disturb the lamellar organization of the newly rejuvenated skin. The negative effects of muscle tension prolong the inflammatory response and increase the metabolic activity during the healing period.[4] From Day 90 to Year 3, BoNTA-injected sides showed smaller FWSS grades than control sides, translating the maintenance of the excellent clinical results of perioral chemabrasion, optimized by BoNTA injections.

Some patients with long rhytides had a better and longer-lasting improvement near the vermilion border when compared to the area near the nose (Figures 4–6). This fact may be explained by the standard protocol of injection sites and doses established for this study. In clinical practice, BoNTA treatment must be individualized. Thus, those patients would need larger doses or a second superior line of injection sites, as suggested by other authors.[10,22]

We agree with Semchyshyn and Sengelmann[10] that when using BoNTA in the perioral area a narrow therapeutic window exists between good results and preservation of mouth functionality. Thus, more than in any other facial area, the optimal recommendation for this specific cosmetic unit is to use BoNTA in combination with other procedures, such as skin ablation techniques and/or fillers.

Conclusions

In severe vertical upper perioral rhytides, the use of adjunctive pretreatment of BoNTA injections in the orbicularis oris muscle is useful in increasing short- and long-term results of perioral chemabrasion. Treatment must be individualized according to each patient's needs.

References

1. Aiache AE. Rejuvenation of the perioral area. Dermatol Clin 1997;15:665–72.
2. Carruthers J, Carruthers A. Aesthetic botulinum A toxin in the mid and lower face and neck. Dermatol Surg 2003;29:486–98.
3. Stagnone JJ. Chemabrasion, a combined technique of chemical peeling and dermabrasion. J Dermatol Oncol 1977;3:217–9.
4. Gassner HG, Sherris DA, Otley CC. Treatment of facial wounds with botulinum toxin A improves cosmetic outcome in primates. Plast Reconstr Surg 2000;105:1948–55.
5. Perkins SW, Sklarew EC. Prevent of facial herpetic infections after chemical peel and dermabrasion: new treatment strategies in the prophylaxis of patients undergoing procedures of the perioral area. Plast Reconstr Surg 1996;98:427–33.
6. Kadunc BV, Di Chiacchio N, Trinidade Almeida AR. Tulle or veil fabric: a versatile option for dressings. J Am Acad Dermatol 2001;47:129–31.
7. Carruthers J, Carruthers A. The adjunctive usage of botulinum toxin. Dermatol Surg 1998;24:1244–7.
8. Carruthers J, Carruthers A. Botox use in the mid and lower face and neck. Semin Cutan Med Surg 2001;20:85–92.

9. Carruthers J, Carruthers A, Maberley D. Deep resting glabellar rhytides respond to BTX-A and Hylan B. Dermatol Surg 2003;29:539–44.

10. Semchyshyn N, Sengelmann RD. Botulinum toxin A treatment of perioral rhytides. Dermatol Surg 2003;29:490–5.

11. Blitzer A, Binder WJ. Current practices in the use of botulinum toxin A in the manegement of facial lines and wrinkles. Facial Plast Surg 2001;9:395–404.

12. Holmkvist KA, Rogers G. Treatment of perioral rhytides. Arch Dermatol 2000;136:725–31.

13. Gin I, Chew J, Rau KA, et al. Treatment of upper lip wrinkles: a comparison of the 950 µsec dwell time carbon dioxide laser to manual tumescent dermabrasion. Dermatol Surg 1999;25: 468–74.

14. Chiarello SE. Commentary: treatment of upper lip wrinkles. a comparison of the 950 µsec dwell time carbon dioxide laser to manual tumescent dermabrasion. Dermatol Surg 1999;25: 473–4.

15. Chew J, Gin I, Rau KA, et al. Treatment of upper lip wrinkles: a comparison of 950 µsec dwell time carbon dioxide laser with unoccluded Baker's phenol chemical peel. Dermatol Surg 1999;25:262–6.

16. Brody HJ. Chemical peeling and resurfacing: history of chemical peels. Philadelphia: BC Decker; 1997. p. 1–5.

17. Harris D, Noodleman FR. Combining manual dermasanding with low strength trichloroacetic acid to improve actinically injured skin. J Dermatol Surg Oncol 1994;20:436–42.

18. Lusthaus S, Benmeir P, Neuman A, et al. The use of sandpaper in chemical peeling combined with dermabrasion of the face. Ann Plast Surg 1993;31:281–2.

19. Kadunc BV, Trinidade Almeida AR. Surgical treatment of facial acne scars based on morphological classification: a Brazilian experience. Dermatol Surg 2003;29:1200–10.

20. West TB, Alster TA. Effect of botulinum toxin type A on movement-associated rhytides following CO_2 laser resurfacing. Dermatol Surg 1999;25:259–61.

21. Zimbler MS, Holds JB, Kokoska MS, et al. Effect of botulinum toxin pretreatment on laser resurfacing results. Arch Facia Plast Surg 2001;3:165–9.

22. Atamoros FP. Botulinum toxin in the lower on third of the face. Clin Dermatol 2003;21:505–12.

Address correspondence and reprint requests to: Bogdana Victoria Kadune, PhD, Rua Gaivota 91, apto 71, São Paulo, Brazil CEP 04522-030, or e-mail: bog@uol.com.br

COMMENTARY

These authors have shown, in a long-term, controlled, and blinded fashion what many cosmetic surgeons who perform perioral resurfacing have long believed: that preoperative injection of botulinum toxin A enhance outcomes. This is an important confirmation of a supposition and also extrapolates to the use of preoperative treatment of other areas to reduce scar formation and enhance outcomes.

STEPHEN MANDY, MD
Miami, FL